

Hybrid fuel: Solid

Hybrid Rocket Engines (HRE)

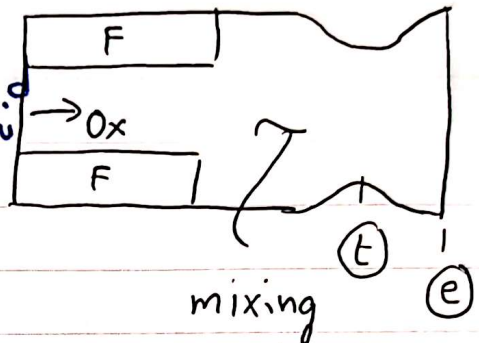
OX: liquid

In the HRE, one propellant is liquid and the other propellant is solid. Typically, the oxidizer is the liquid and the fuel is the solid. HRE with liquid fuel and solid oxidizer are referred to as reverse hybrids.

Advantages:

- ① Safer handling
- ② Throttling capability (start, stop, restart)
- ③ Allows use of variety of fuels and oxidizers
- ④ Reduced temperature sensitivity
- ⑤ Higher I_{sp} than SRM, and higher bulk density than LPRE
- ⑥ While cracks and voids in the solid result in unpredictable operation, they are not catastrophic as in SRM.

Reverse hybrid
OX: Solid
fuel: liquid



Disadvantages:

- ① Oxidizer-fuel ratio changes during burn
- ② Low regression rate (an order of magnitude smaller than the solid propellant). This results in complicated fuel geometry to generate significant thrust. This also offsets the bulk density advantage to some extent.
- ③ Lower combustion efficiency.

In the typical operating condition of the hybrid rocket engine, the regression rate (r) of the fuel is empirically found to be governed by the mass flux (G) $\left\{ G \stackrel{\text{def}}{=} \frac{\dot{m}}{A} \right\}$

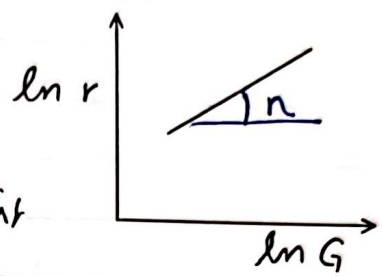
solid rockets $\rightarrow r = a p_o^n$

$$r = a' G^{n'}$$

where a' and n' are constants.

Since G includes both fuel and oxidizer mass fluxes $\{G = (G_f + G_{ox})\}$, it is more convenient to use

$$G = \frac{\dot{m}}{A}$$



$$r = a G_{ox}^n \quad (1)$$

with a and n empirically found.

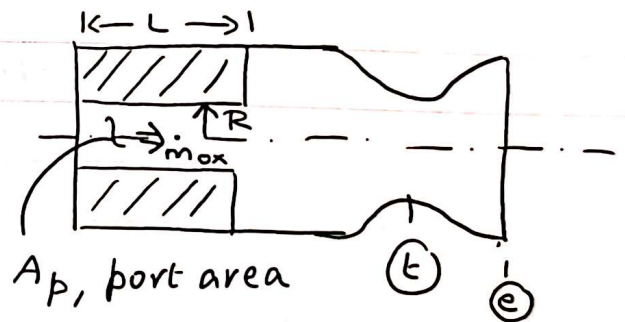
$$\dot{m} = \rho_f r A_b = \rho_f a G_{ox}^n A_b \quad \{A_b - \text{fuel burn area}\}$$

Consider the simple cylindrical arrangement.

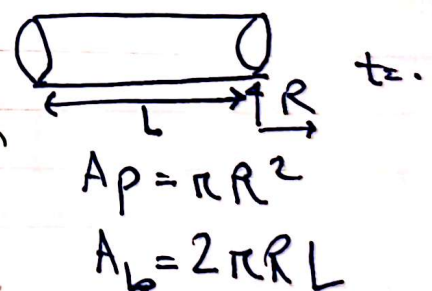
$$A_p = \pi R^2, \quad A_b = 2\pi R L$$

$$r = a G_{ox}^n = a \left(\frac{\dot{m}_{ox}}{A_p} \right)^n$$

$$= a \left(\frac{\dot{m}_{ox}}{\pi R^2} \right)^n$$



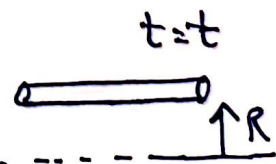
$$\frac{dR}{dt} = r = a \left(\frac{\dot{m}_{ox}}{\pi R^2} \right)^n = a \frac{\dot{m}_{ox}^n}{\pi^n} R^{-2n}$$



$$R^{2n} dR = a \left(\frac{\dot{m}_{ox}}{\pi} \right)^n dt$$

Integrate from $t = 0 \{R = R_i\}$ to $t = t \{R = R\}$:

$$\frac{R^{2n+1} - R_i^{2n+1}}{(2n+1)} = a \left(\frac{\dot{m}_{ox}}{\pi} \right)^n t$$



$$\Rightarrow R = \left[a(2n+1) \left(\frac{\dot{m}_{ox}}{\pi} \right)^n t + R_i \right]^{\frac{1}{2n+1}} \quad (2)$$

$$\dot{m}_f = \rho_f A_b r = \rho_f (2\pi R L) a \left(\frac{\dot{m}_{ox}}{\pi R^2} \right)^n \quad (3)$$

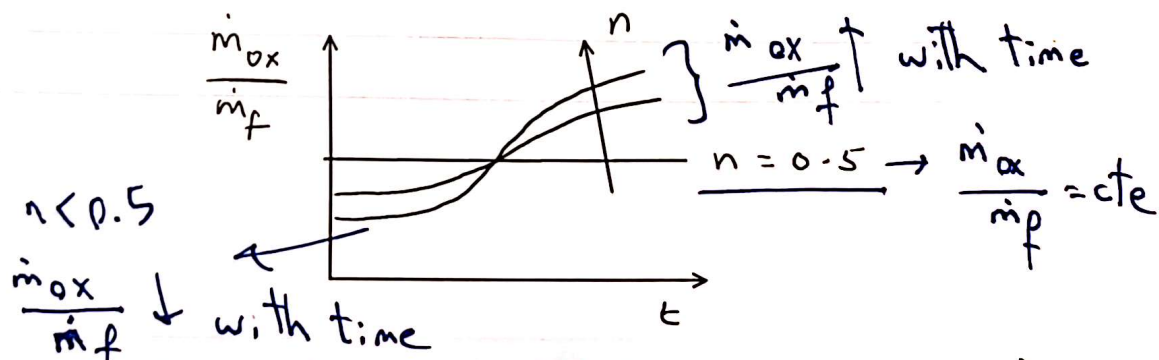
Substituting for R from (2) in (3),

$$\dot{m}_f = 2a\pi^{1-n} \rho_f L \dot{m}_{ox}^n \left[a(2n+1) \left(\frac{\dot{m}_{ox}}{\pi} \right)^n t + R_i \right]^{\frac{1-2n}{2n+1}} \quad (4)$$

Observe that $\dot{m}_f$ is constant if $n = 0.5$

$$r = \frac{\dot{m}_{ox}}{\dot{m}_f} = \frac{1}{2\rho_f a L} \left(\frac{\dot{m}_{ox}}{\pi} \right)^{1-n} \left[a(2n+1) \left(\frac{\dot{m}_{ox}}{\pi} \right)^n t + R_i \right]^{\frac{2n-1}{2n+1}}$$

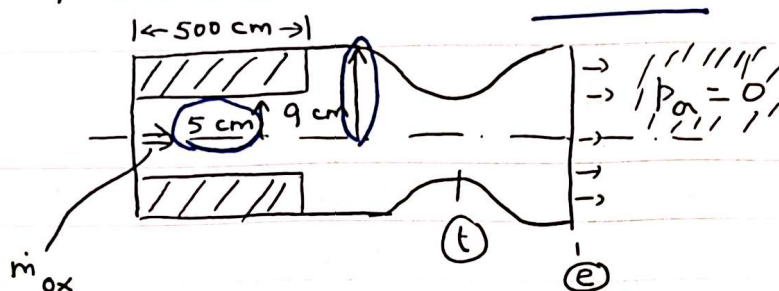
If $n > 0.5$, $\dot{m}_f$ decreases with time. Assuming $\dot{m}_{ox}$ is constant, $\left(\frac{\dot{m}_{ox}}{\dot{m}_f} \right)$ will increase with time.



To generate large thrust, multiple ports are required (recall the low regression rates of hybrid rocket propellants). An example, seven-port ($N=7$) configuration is shown below.

Example

A small-scale HRE with cylindrical fuel geometry is designed for operation in vacuum ($p_a = 0$).



The fuel is HTPB and oxidizer is LOX. The regression rate of HTPB in oxygen stream is known to be

$$r = 1.6086 G_{ox}^{0.681}, \text{ where}$$

G_{ox} is in $\frac{kg}{cm^2 \cdot s}$ and r is in cm/s . The mass flow

rate of the oxidizer ($\dot{m}_{ox}$) is constant at 6.283 kg/s .

The HRE generates an initial specific impulse of 250 s .

The burn characteristics are given in the following table for HTPB - oxygen reaction*.

→ convert to m/s

$(\dot{m}_{ox}/\dot{m}_f)$	$c^* \text{ (ft/s)}$	γ	$\bar{M}_{prod}^\#$
1.0	4825	1.308	18
1.2	5180	1.282	19
1.4 → 1.52	5543	1.239	20
1.6	5767	1.201	20
1.8	5882	1.171	21
2.0	5912	1.152	21
2.2	5885	1.143	22
2.4	5831	1.138	23
2.6	5768	1.135	25
2.8	5703	1.133	26
3.0	5639	1.132	27

$\uparrow$ $\frac{\dot{m}_{ox}}{\dot{m}_f}$ with time

* Table 16.2, "Rocket Propulsion Elements," Sutton, Biblarz

† This data column is not from Table 16.2 of Sutton, et al, text.

- What is the initial oxidizer flux ($G_{ox,i}$)?
- Find the initial fuel regression rate (r_i) and the initial fuel mass flow rate ($\dot{m}_{f,i}$) $\{SG_f = 0.915\}$
- What is the initial ox-f ratio ($\dot{m}_{ox,i}/\dot{m}_{f,i}$)?
- What is the initial thrust generated?
- What are the initial values of c^* and C_f ?
- Calculate the initial value of the flame temperature.
- What is the nozzle area ratio (A_e/A_t) and the exit Mach number (M_e)?
- If the initial chamber pressure is 1.5 MPa, what is the throat area (A_t) of the nozzle?
- What is the HRE's burn time?

$$a) G_{ox} = \frac{\dot{m}_{ox}}{A_p} = \frac{\dot{m}_{ox}}{\pi R^2} \quad (A_p - \text{port area})$$

$$G_{ox,i} = \frac{\dot{m}_{ox,i}}{\pi R_i^2} = \frac{6.283}{\pi (5)^2} = 0.08 \frac{\text{kg}}{\text{cm}^2 \cdot \text{s}}$$

$$b) r_i = a G_{ox,i}^{0.681} = 1.6086 (0.08)^{0.681} = 0.288 \text{ cm/s}$$

$$\begin{aligned} \dot{m}_{f,i} &= \rho_f A_{b,i} r_i = \rho_f 2\pi R_i L r_i \\ &= (0.915) (2) (\pi) (5) (500) (0.288) \\ &= 4139 \text{ g/s} = 4.139 \text{ kg/s} \end{aligned}$$

$$c) \frac{\dot{m}_{ox,i}}{\dot{m}_{f,i}} = \frac{6.283}{4.139} = 1.52$$

$$d) T = \dot{m} u_{eq} = \dot{m} I_{sp} g_e$$

$$T_i = \dot{m}_i I_{sp,i} g_e$$

$$\dot{m}_i = (\dot{m}_{ox,i} + \dot{m}_{f,i}) = (6.283 + 4.139) = 10.422 \text{ kg/s}$$

Assume $\frac{A_e}{A_t} \rightarrow$ get $M > 1 \rightarrow$ get $\frac{P_e}{P}$ check C_T

$$\tau_i = (10.422)(250)(9.81) = 25560 \text{ N}$$

e) From the data table, at $\left(\frac{\dot{m}_{ox}}{\dot{m}_f}\right) = 1.52$, (interpolate data)

$$c_i^* = 5677 \text{ ft/s} = 1731 \text{ m/s}, \gamma_i = 1.22, \bar{M}_i = 20$$

$$u_{eq} = C_T c^* \left\{ u_{eq} = \frac{\tau}{\dot{m}} = \left(\frac{\tau}{p_0 A_t} \right) \left(\frac{p_0 A_t}{\dot{m}} \right) = C_T c^* \right\} = u_{eq}$$

$$\Rightarrow C_{T,i} = \frac{u_{eq,i}}{c_i^*} = \frac{I_{sp,i} g_e}{c_i^*} = \frac{(250)(9.81)}{1731} = 1.42$$

$$f) c^* = \sqrt{\frac{1}{\gamma} \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{\gamma-1}} \frac{\bar{R} T_0}{\bar{M}}}$$

$$c_i^* = \sqrt{\frac{1}{\gamma_i} \left(\frac{\gamma_i+1}{2} \right)^{\frac{\gamma_i+1}{\gamma_i-1}} \frac{\bar{R} T_{0,i}}{\bar{M}_i}} \Rightarrow T_{0,i} = 3068 \text{ K}$$

$\frac{8314.3 \text{ J}}{\text{kmol} \cdot \text{K}}$

$1731 \frac{\text{m}}{\text{s}}$

$$g) C_T = \left\{ \frac{2\gamma^2}{\gamma-1} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \left[1 - \left(\frac{p_e}{p_0} \right)^{\frac{\gamma-1}{\gamma}} \right] \right\}^{\frac{1}{2}} + \frac{p_e - p_a}{p_0} \left(\frac{A_e}{A_t} \right)$$

Initially, $C_T = C_{T,i} = 1.42, \gamma = \gamma_i = 1.22$

Assume a value for $\frac{A_e}{A_t}$. Solve for $M_e (> 1)$. Find $\left\{ \frac{A_e}{A_t} ? \right.$
 corresponding $\left(\frac{p_e}{p_0} \right)$. Calculate C_T . Repeat until $C_T = 1.42$. $\left. \frac{P_e}{P} ? \right.$

$$\Rightarrow \frac{A_e}{A_t} = 1.67, M_e = 1.894$$

$$h) C_{T,i} = \frac{\tau_i}{p_{0,i} A_t} \Rightarrow p_{0,i} A_t = \frac{\tau_i}{C_{T,i}} = \frac{25560}{1.42} = 18000 \text{ N}$$

$$p_{0,i} = 1.5 \text{ MPa} \Rightarrow A_t = \frac{18000}{1.5(10^6)} = 0.012 \text{ m}^2 = 120 \text{ cm}^2$$

$$\{ R_c = 6.18 \text{ cm} \}$$

$$(i) R = \left\{ a(2n+1) \left(\frac{\dot{m}_{ox}}{\pi} \right)^n t + R_i^{2n+1} \right\}^{\frac{1}{2n+1}}$$

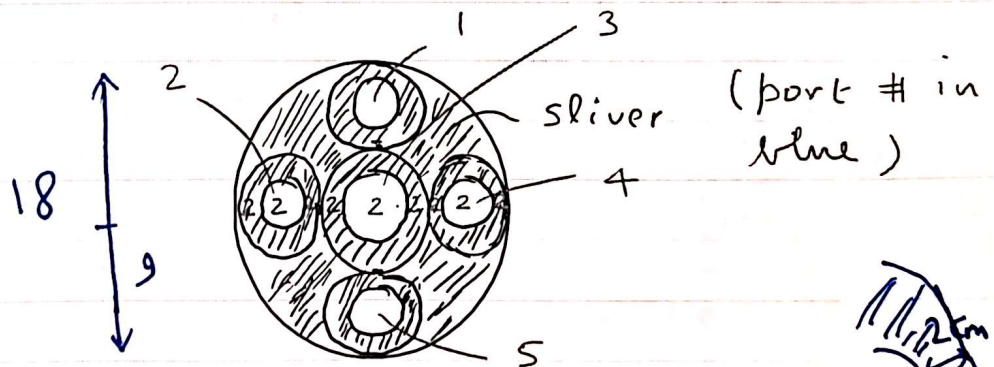
$$R_f = \left\{ 1.6086 (2.362) \left(\frac{6.283}{\pi} \right)^{0.681} t_b + 5^{2.362} \right\}^{\frac{1}{2.362}}$$

9

$$\Rightarrow t_b = 22.1 \text{ s}$$

500 cm

Comment : Observe the length of the HRE. It is 500 cm (5 m = 16.4 ft). The thrust is not large 25560 N, which is about 5746 lbf. To reduce the length (or generate larger thrust for a given length), multiple ports are utilized. Reconsider the previous example of the HRE, with the same outside radius of 9 cm, but with five ports of equal cross-sections, see Figure, below.



Since all the ports have the same cross-section, each port has an outside radius of 3 cm. If the initial radius of each port is 1 cm, then the fuel ring thickness in each port is 2 cm.



If the oxidizer flow is equally split, the following values represent quantities per port :

...

values represent quantities per port:

$$\dot{m}_{ox} = \frac{6.283}{5} = \boxed{1.257 \frac{\text{kg}}{\text{s}}}$$

$$G_{ox,i} = \frac{\dot{m}_{ox}}{\pi R_i^2} = \frac{1.257}{\pi (1^2)} = \boxed{0.4 \frac{\text{kg}}{\text{cm}^2 \cdot \text{s}}}$$

$$r_i = 1.6086 (0.4)^{0.681} = \boxed{0.862 \frac{\text{cm}}{\text{s}}}$$

$$m_{f,i} = \rho_f \underbrace{2\pi R_i L}_{A_{b,i}} \underbrace{r_i}_{G_{ox,i}} = (0.915) (2) (\pi) (1) (0.862) L = \boxed{4.956 L \text{ (g/s)}}$$

$$\frac{\dot{m}_{ox,i}}{\dot{m}_{f,i}} = \frac{1.257 \times 10^3 \frac{\text{g}}{\text{s}}}{4.956 \times 167 \frac{\text{g}}{\text{s}}} = \boxed{1.51}$$

Since the total initial fuel flow rate is 4139 g/s [part (b), previous example],

$$\dot{m}_{f,i} \text{ per port} \leftarrow 4.956 L = \frac{\dot{m}_{f,i} \text{ total from previous Ex}}{5} \Rightarrow \boxed{L = 167 \text{ cm}}$$

Observe the two-thirds reduction in length from 500 cm to 167 cm

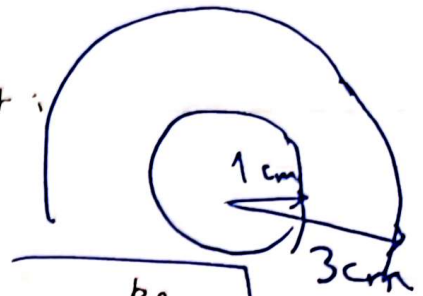
$$G_{ox,final} = \frac{\dot{m}_{ox}}{\pi R_f^2} = \frac{1.257}{\pi (3^2)} = \boxed{0.0445 \frac{\text{kg}}{\text{cm}^2 \cdot \text{s}}}$$

$$r_f = 1.6086 (0.0445)^{0.681} = \boxed{0.193 \frac{\text{cm}}{\text{s}}}$$

$$\dot{m}_{f,final} = \rho_f \underbrace{2\pi R_{f,final} L}_{A_{b,final}} \underbrace{r_{f,final}} = (0.915) (2) (\pi) (3) (167) (0.193) = 555.9 \frac{\text{g}}{\text{s}} = \boxed{0.556 \frac{\text{kg}}{\text{s}}}$$

$$\frac{\dot{m}_{ox,f}}{\dot{m}_{f,f}} = \frac{1.257}{0.556} = \boxed{2.26}$$

From the data table, at $\frac{\dot{m}_{ox,f}}{\dot{m}_{f,f}} = \boxed{2.26}$,



$$C_{final}^* = 5869 \text{ ft/s} = 1789 \text{ m/s}, \gamma = 1.14, \bar{M} = 22.3$$

$$T_{0,f} = 3479 \text{ K}$$

$$\frac{A_e}{A_t} = 1.67, \gamma = 1.14 \Rightarrow M_e = 1.85, C_{T,f,ina} = 1.42$$

$$u_{eq,final} = C_f^* C_{T,f} = (1789)(1.42) = 2540 \text{ m/s}$$

$$I_{sp,f} = \frac{u_{eq,f}}{g_e} = \frac{2540}{9.81} = 259 \text{ s}$$

$$R_f = \left\{ a(2n+1) \left(\frac{\dot{m}_{ox}}{\pi} \right)^n t_b + R_i^{2n+1} \right\}^{\frac{1}{2n+1}}$$

$$\underline{3} = \left\{ 1.6086 (2.362) \left(\frac{1.257}{\pi} \right)^{0.681} t_b + 1^{2.362} \right\}^{\frac{1}{2.362}}$$

$$\Rightarrow t_b = \underline{6.1 \text{ s}} \text{ (much less than the } \underline{22.1 \text{ s}} \text{ with single cylindrical fuel geometry)}$$

